# SHEAVES AND MODULES AS FILTERED COLIMITS OF STANDARD PRESENTATIONS BUILT FROM A BASIS OF QUASI-COMPACT OBJECTS

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Both the statement and the proof contain commutative diagrams typeset with Xy-pic; the retained two-slice preamble of this source supplies the native Xy-pic setup, so they compile as in the source.

*Editorial note.* The counted span cites a situation and a quotient lemma, both recorded in assembly.external_references with canonical tags as navigation pointers.

*Allowed standard prerequisites (editorial).*

- Every sheaf of sets on a site with enough quasi-compact objects is a quotient of a coproduct of representables. Stacks Project, Modules on Sites, Lemma lemma-module-quotient-direct-sum (tag 09B3). Produces the surjection and the fibre product presentation used to build the coequalizer.
- The quasi-compact objects situation. Stacks Project, Modules on Sites, Situation situation-quasi-compact-objects (tag 09AZ). Supplies the basis B and the quasi-compactness hypotheses that make the finite cutting argument work.

*Source reference.* Original reference 0937: sources/stacks-chow/tags.txt, source lines 11505–11505. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 0938: sources/stacks-chow/tags.txt, source lines 11506–11506. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 0939: sources/stacks-chow/tags.txt, source lines 11507–11507. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original lemma 093B: sources/stacks-chow/tags.txt, source lines 11509–11509. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 0738: sources/stacks-chow/tags.txt, source lines 9052–9052. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original lemma 0936: sources/stacks-chow/tags.txt, source lines 11504–11504. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *In Situation 0937 assume (0938) and (0939) hold.*

*Date*: stacks-project:sites-modules.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *Every sheaf of sets is a filtered colimit of sheaves of the form*

$$\text{Coequalizer}\left( \coprod_{j=1,\ldots,m} h^\#_{V_j} \longrightarrow \coprod_{i=1,\ldots,n} h^\#_{U_i} \right) \tag{0.1.1}$$

*with $U_i$ and $V_j$ in $\mathcal{B}$.*

(2) *If $\mathcal{O}$ is a sheaf of rings, then every $\mathcal{O}$-module is a filtered colimit of sheaves of the form*

$$\text{Coker}\left( \bigoplus\nolimits_{j=1,\ldots,m} j_{V_j!}\mathcal{O}_{V_j} \longrightarrow \bigoplus\nolimits_{i=1,\ldots,n} j_{U_i!}\mathcal{O}_{U_i} \right) \tag{0.1.2}$$

*with $U_i$ and $V_j$ in $\mathcal{B}$.*

*Proof.* Proof of (1). By Lemma 093B every sheaf of sets $\mathcal{F}$ is the target of a surjection whose source is a coprod $\mathcal{F}_0$ of sheaves the form $h^\#_U$ with $U \in \mathcal{B}$. Applying this to $\mathcal{F}_0 \times_{\mathcal{F}} \mathcal{F}_0$ we find that $\mathcal{F}$ is a coequalizer of a pair of maps

$$\coprod_{j\in J} h^\#_{V_j} \longrightarrow \coprod_{i\in I} h^\#_{U_i}$$

for some index sets $I$, $J$ and $V_j$ and $U_i$ in $\mathcal{B}$. For every finite subset $J' \subset J$ there is a finite subset $I' \subset I$ such that the coproduct over $j \in J'$ maps into the coprod over $i \in I'$ via both maps, see Sites, Lemma 0738. (Details omitted; hint: an infinite coproduct is the filtered colimit of the finite sub-coproducts.) Thus our sheaf is the colimit of the cokernels of these maps between finite coproducts.

Proof of (2). By Lemma 093B every module is a quotient of a direct sum of modules of the form $j_{U!}\mathcal{O}_U$ with $U \in \mathcal{B}$. Thus every module is a cokernel

$$\text{Coker}\left( \bigoplus\nolimits_{j\in J} j_{V_j!}\mathcal{O}_{V_j} \longrightarrow \bigoplus\nolimits_{i\in I} j_{U_i!}\mathcal{O}_{U_i} \right)$$

for some index sets $I$, $J$ and $V_j$ and $U_i$ in $\mathcal{B}$. For every finite subset $J' \subset J$ there is a finite subset $I' \subset I$ such that the direct sum over $j \in J'$ maps into the direct sum over $i \in I'$, see Lemma 0936. Thus our module is the colimit of the cokernels of these maps between finite direct sums. □